MailSentra

Multi-Modal Email Threat Detection & Defense Platform

Thesis Document

COMMON INTEGRATED SOLUTION REPORT

Sri Lanka Institute of Information Technology

Sri Lanka

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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration and supervisor certification must be completed by the actual authors and supervisor before submission. The wording below is provided for review and signature; this draft does not assert that signatures or approvals have been obtained.

We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, while retaining the right to use its content in future work.

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ABSTRACT

MailSentra is an integrated email-security research system intended to combine scam detection, steganographic image analysis, protective response and contextual user awareness within the Thunderbird email client. The design addresses the separation between detecting suspicious content and helping users understand and respond to it. After an email arrives, the user explicitly requests analysis. A text-analysis component combines a DistilBERT scam probability with scam-related indicators through a trained logistic model and calibration, while reporting a separate AI-writing assessment. An independent image-analysis component classifies supported attachments as clean or steganographic and produces Grad-CAM visual explanations. A defense and awareness component maps findings to protective actions, selects evidence-consistent explanation templates and maintains a transparent interaction-based security profile. A fixed severity matrix combines text and image findings into one response. This report synthesises the supplied architecture and design documents, defines the integration workflow and presents an evaluation framework for classification quality, calibration, enforcement, usability and operational cost. The available evidence supports a coherent architectural proposal but does not establish measured detection accuracy, resource efficiency, user-learning gains or successful end-to-end enforcement. Those outcomes require reproducible experiments and implementation records. The principal contribution described here is the integration of separate text and image assessments with understandable, policy-driven responses. The draft identifies unresolved interface and policy questions and outlines the evidence needed to complete a defensible final evaluation.

Keywords: email security, scam detection, steganalysis, explainable AI, user awareness

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No experimental screenshots or measured charts were supplied. Architecture and evaluation tables are provided in the main text. Final figures should be inserted with chapter-based captions after the integrated deployment is confirmed.

LIST OF TABLES

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
C1 / C2 / C3	Component 01 / Component 02 / Component 03
DKIM	DomainKeys Identified Mail
DMARC	Domain-based Message Authentication, Reporting, and Conformance
DNS	Domain Name System
EML	Email message file format
Grad-CAM	Gradient-weighted Class Activation Mapping
LSB	Least significant bit
PNG	Portable Network Graphics
SDG	Sustainable Development Goal
SME	Small and medium-sized enterprise
SPF	Sender Policy Framework
TF-IDF	Term frequency–inverse document frequency
URL	Uniform resource locator

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Email security depends on more than classifying the words in a message. Users also encounter sender identities, authentication results, links and attachments, and must decide what to do when a warning appears. MailSentra addresses this interaction by connecting detection with protective action and contextual explanation. Its intended users are individuals and small and medium-sized enterprises that need an understandable email-security workflow [1].

The supplied overview motivates the project through concerns about fragmented tools, false positives, affordability and awareness that is disconnected from the message being examined. These statements establish the team's motivation; they are not comparative findings. Claims that existing products fail to stop a particular percentage of scams, or that MailSentra achieves fewer false positives, require independently sourced evidence and controlled comparison before inclusion as conclusions.

Artificial intelligence creates two distinct questions in this study: whether a message is a scam and whether its wording resembles AI-generated writing. The architecture deliberately separates these questions. A legitimate AI-assisted message should not receive a higher scam score merely because of its writing style, while a human-written phishing message must remain detectable [2].

The research also considers image attachments that may contain hidden data. A steganalysis prediction is a separate assessment of an image, not a direct measurement of scam intent. Connecting it with text findings expands the evidence available to the response layer, but does not make either classifier infallible.

1.2 Stakeholders and user needs

The primary stakeholder is the Thunderbird user who selects an email for analysis. This user needs a clear outcome, an explanation tied to the selected message and accurate reporting of whether a protective action actually occurred. A second

stakeholder is the operator who installs and maintains the extension and its analysis services. That role needs understandable configuration, health information and recovery instructions.

The academic team and supervisors need traceable research questions, reproducible experiments and clear ownership of component and integration work. Potential organisational adopters would also need evidence about privacy, support requirements and total operating cost. No interviews, participant records or signed stakeholder requirements were supplied. The needs described here are derived from the design documents and must be validated through stakeholder engagement.

1.3 Alignment with the Sustainable Development Goals

The project proposes a connection with SDG 4 through contextual digital-security learning and with SDG 9 through accessible security tooling. This is an intended contribution, not a measured social outcome. The report should ultimately connect any selected target to an observable outcome, such as a validated improvement in understanding or a documented reduction in deployment barriers.

Any claimed benefit must be balanced against possible negative effects: false warnings can interrupt legitimate work, excessive confidence in a heatmap can mislead users, and destructive enforcement can remove legitimate communication. Responsible evaluation must therefore measure both benefits and costs. Formal SDG target references and impact evidence remain to be added.

1.4 Purpose and scope of the common report

This common report explains MailSentra as one integrated system. It covers the overall problem, component responsibilities, shared workflow, integration contracts, implementation requirements, evaluation and responsible adoption. Detailed component experiments belong in the individual research reports and should be cross-referenced when available.

The supplied common-report template assumes four research components and four team members [5]. The system overview instead specifies three research components [1]. This draft follows the documented three-component system. Shared orchestration

is described as integration infrastructure; it is not assigned the status of a fourth research contribution or a student owner without confirmation.

2 OVERALL PROBLEM DEFINITION AND THREE SUBPROBLEMS

2.1 Overall problem definition

The common problem is the difficulty of giving an email user a coherent, understandable response when suspicious message text or image attachments are encountered. A detection label alone does not tell the user which evidence matters, which actions remain available or how to handle similar messages later. The project investigates an integrated workflow connecting separate analyses to a single protective response and explanation.

The problem is defined at the level of the user's decision and the system's response. AI authorship is a contextual attribute rather than a necessary condition for a scam. Similarly, hidden data in an image does not establish a malicious payload. These distinctions prevent the project from treating a writing-style estimate or steganalysis class as proof of criminal intent.

2.2 Root causes and overall research gap

The design documents identify a gap between detection, intervention and learning. Text classifiers can provide a score without explaining operational consequences. Attachment analysis can produce a separate alert that the user must reconcile. General awareness material may not explain the specific email under review. MailSentra proposes one event-based workflow that relates these outputs [1], [3].

This is a design gap identified by the supplied project material. A defensible literature gap still requires a systematic review of comparable systems, their deployment assumptions, explanation methods, enforcement capabilities and evaluations. It is premature to claim that no existing product combines these functions. Chapter 7 therefore frames differentiation as a hypothesis for validation.

2.3 Subproblem and component mapping

The first subproblem is assessing scam-related risk from message content and sender, URL and attachment indicators while keeping AI-writing assessment separate. The

second is converting findings into a consistent action, an understandable explanation and a transparent interaction profile. The third is assessing supported images for steganographic patterns and explaining the model's prediction.

The components are operationally related but retain separate responsibilities. C1 and C3 analyse different evidence and should not condition their execution on one another's positive finding. C2 receives their outputs and owns response policy. The extension and shared integration layer connect these responsibilities to the email selected by the user. Table 2.1 records this division.

Table 2.1 Research subproblems and component responsibilities

Component	Input	Method and output
C1	Selected email	Text and indicator analysis; scam score, evidence, status and AI-writing assessment
C2	C1 and/or C3 findings	Policy response, awareness explanation and interaction profile
C3	Supported image attachments	Image classifier; prediction, class scores and Grad-CAM

2.4 Overall aim and group objectives

The overall aim is to design and evaluate an integrated Thunderbird workflow that combines independent email-text and image assessments with understandable, policy-driven defense and contextual awareness.

The group objectives are to preserve separate scam and AI-writing outputs; support independent image analysis; produce one deterministic response for each completed analysis event; ground explanations in recorded evidence; record user interactions transparently; and evaluate detection, enforcement, usability and resource use. Each objective needs an associated acceptance criterion and reproducible evidence. Numerical targets cannot be retrospectively invented from results and must be agreed before evaluation.

2.5 Scope assumptions and constraints

The system is scoped to Thunderbird and user-selected emails after normal inbox delivery. It is not described as a mail-server gateway that intercepts all incoming mail. Text and supported image attachments are in scope; audio and video analysis are excluded. The overview limits image detection to PNG images and LSB-related steganography, whereas the C3 summary also mentions JPEG. PNG is treated as the conservative documented scope until this conflict is resolved.

The documents do not establish deployment hardware, final service endpoints, actual storage encryption, dataset provenance or the exact model used for C3. Those details must be confirmed through implementation evidence. No measured accuracy, latency, memory use or user-learning outcome is inferred from the presence of a model or dashboard.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

MailSentra provides a selected-message analysis workflow in Thunderbird. The user invokes “Analyze this email”; the selected message is converted to a standard representation and supplied to independent text and image paths. Results are correlated to the same event and passed to the defense and awareness layer, which produces one action and one evidence-linked explanation [1]–[4].

The central value proposition is coherence: the user should see which message was analysed, what was found, what the system attempted and what actually succeeded. An action recommendation is not equivalent to Thunderbird enforcement. A complete implementation must distinguish those states and retain the evidence needed to review them.

3.2 High level architecture

The logical architecture contains a Thunderbird extension, shared ingestion and orchestration, C1 text and indicator analysis, C3 image analysis, C2 defense and awareness, and an event store feeding a dashboard. C1 and C3 operate as sibling analysis paths. The overview and C2 specification support independent delivery to C2; the C1 diagram also depicts external C3 evidence in its output bundle. This draft uses the sibling interpretation and records the discrepancy for confirmation.

An architectural drawing should separate processing responsibilities from deployment boundaries. The supplied sources do not fully specify hosting or storage. Consequently, this draft does not convert processing boxes into asserted network services. Table 3.1 gives the logical data-flow relationships that a final deployment diagram must preserve.

Table 3.1 Logical architecture and data flow

Source	Destination	Information
User / Thunderbird	Shared ingestion	Selected-message analysis request

Source	Destination	Information
Ingestion	C1	Normalised message
Ingestion	C3	Supported image attachments
C1 and C3	C2	Independent analysis outputs
C2	Extension	Action and explanation
Extension	Event record	Actual enforcement and user response
Event record	Dashboard / profiler	Activity and interaction history

3.3 Component 1 summary

C1 ingests EML, raw message, JSON or text inputs and creates a normalised object containing subject, body, sender, headers, links and attachment metadata. Its scam-related lane performs sender-authentication, domain, social-engineering, URL and attachment-risk checks alongside DistilBERT classification. DistilBERT is a compact distilled language-model architecture; its published general-language results are not MailSentra's detection results [2], [7].

The architecture computes $z = b + w_m p_m + \Sigma w_i x_i$, followed by $p_{combined} = 1/(1 + \exp(-z))$ and score = 100 × Calibrate($p_{combined}$). The coefficients must be learned from labelled data rather than selected for presentation. The rule-feature definitions, training split, fitted coefficients and calibration procedure are not supplied and therefore remain requirements for reproducibility.

The AI-writing lane uses distilgpt2 predictability and stylometric features, including sentence-length variation and repetition, with a small classifier. Its result is reported separately. C1 supplies scam risk, supporting evidence, analysis status and AI-authorship assessment. The minimum-text gate for AI-writing must be distinguished from the diagram's broader early-exit gate.

3.4 Component 2 summary

C2 contains a Defense Engine, Awareness Engine and Vulnerability Profiler. The dominant policy in the overview and later C2 sections maps the scam score directly to a severity band; evidence strength, completeness and AI authorship inform

explanations. Earlier passages describe evidence as influencing the decision, so the final implementation must establish which policy is authoritative [1], [3].

The awareness design uses TF-IDF and a linear classifier to select a predefined category or psychological-trigger template. An allow-list and evidence-consistency check constrain selection; deterministic priority rules provide fallback. The classifier does not generate arbitrary explanatory prose. Profile evolution is calculated by documented rules rather than an opaque behavioural model.

The specified score bands are Low 0–10, Medium 11–39, High 40–69 and Critical 70–100. They map to normal access, restricted access, quarantine and blocking respectively. The treatment of fractional scores between integer boundaries must be specified explicitly. This draft does not silently choose a rounding policy.

Table 3.2 Specified score bands and actions

Score band	Severity	Action
0–10	Low	Allow
11–39	Medium	Restrict
40–69	High	Quarantine
70–100	Critical	Block / permanent removal as specified

3.5 Component 3 summary

C3, named StegoGuard in its overview, identifies supported image attachments and submits their pixels to a trained classifier through a Flask backend. It processes multiple supported attachments separately. For each image it returns Clean or Stego, clean and stego class scores, a Grad-CAM heatmap and image metadata [4].

Grad-CAM uses gradient information to produce a class-discriminative visualisation of influential image regions [8]. It is an explanation of model behaviour, not an extracted payload, exact embedding map or forensic proof. C3’s classifier remains probabilistic and may produce false positives or false negatives. The supplied summary does not identify its architecture, training data or measured performance.

3.6 Shared integration infrastructure

The fourth-component placeholder in the template is adapted here to shared infrastructure because only three research components are documented. The integration layer coordinates message selection, normalisation, correlation, analysis completion and delivery of results to C2. The extension translates a response into Thunderbird operations when the user has granted the necessary permissions.

For clean results, the C2 document proposes a lightweight scan record without invoking full defense and awareness processing. This keeps scan accounting distinct from the flagged-event feed. The exact shared-log implementation and access rules must be demonstrated before being described as deployed functionality.

3.7 End to end user and data flow

The email first arrives through normal Thunderbird delivery. The user selects it and requests analysis. The extension captures only the selected content needed for the workflow. C1 normalises and analyses the message, while C3 independently determines whether supported image attachments are available and analyses each eligible image.

If either analysis produces a positive finding, C2 interprets the available result or combines both severities using the fixed matrix. It determines an action, selects an evidence-consistent explanation and makes the event available to the dashboard. Recorded user responses contribute to the profile when sufficient history exists. A fully clean completed scan produces a quiet scan record rather than a threat alert.

A missing, failed or skipped analysis must not be treated as a clean prediction. The user-facing result should preserve those states. The mechanism for waiting for both paths, timing out a slow path and revising an event after late completion is not defined in the sources and requires an explicit integration contract.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

An interface-first strategy is appropriate because each research component produces different evidence. The team should agree the message identity, event identity, result envelopes and allowed status values before integrating enforcement. This is a proposed integration method; repository records and test reports are required to establish the method actually used.

Integration should proceed from synthetic fixture messages to read-only analysis and then to protective operations on disposable test mailboxes. This ordering makes response decisions observable before introducing destructive effects. Each integration checkpoint should retain the relevant code revision, model version, configuration and expected outcome.

4.2 Interfaces data contracts and shared models

The conceptual contract requires a correlation identifier shared across C1, C3, C2 and enforcement. C1 supplies a bounded scam score when scoring ran, evidence items with provenance, completion status and a separate AI-writing assessment. C3 supplies a per-attachment result and image identity. C2 supplies the policy result, explanation and action request.

Table 4.1 specifies the minimum conceptual fields. It is a proposed contract rather than an extracted live API schema. The final thesis should replace or supplement it with verified request and response schemas, supported versions, sample payloads and field-level validation tests.

Score scale must be explicit. A C1 score described on a 0–100 scale must not be confused with a C3 output on a 0–1 scale. Missing values should use a documented unavailable state, not zero. Prediction, model score and calibrated probability should be named according to what they actually represent.

Table 4.1 Proposed conceptual integration contract

Producer	Required content	Validation need
Shared layer	Event and message identifiers	Correlation and retry semantics
C1	Score, evidence, status, AI assessment	Scale, range, missing-value semantics
C3	Attachment ID, verdict, scores, heatmap	Supported type, scale and per-image status
C2	Severity, policy, explanation, action	Policy version and evidence consistency
Extension	Requested and executed action	Permission result and mailbox outcome

4.3 Shared nonfunctional requirements

The term lightweight must be operationalised through latency, memory, processor utilisation, startup time, installation size and hardware requirements. The use of compact models motivates measurement but does not itself demonstrate low overhead. Usability requires users to distinguish a result, its uncertainty and the action actually executed.

Security and privacy requirements include limiting analysis to selected content, minimising retention, validating untrusted message and image input, protecting configuration and controlling access to stored records. Storage encryption is mentioned in the C2 architecture but is not demonstrated by the supplied material. Accessibility, maintenance and reliability also need testable criteria rather than general assurances.

4.4 Integrated workflow and exception handling

C2’s four-by-four severity matrix usually returns the higher severity, with High plus High escalating to Critical. It applies when both analysis paths provide applicable findings. A single active finding follows its own mapped severity. Table 4.2 preserves the specified matrix without converting it into a learned fusion model.

Exception handling must cover unsupported images, corrupt attachments, unavailable models, analysis timeouts, missing headers, insufficient text and refused Thunderbird permissions. A failed analysis should produce a visible incomplete

status. A failed mailbox operation should retain the proposed action and the enforcement failure separately, allowing the user to understand what remains accessible.

The documents specify permanent removal for Critical findings, while quarantine has a review or release path in the overview. Before destructive behaviour can be claimed as complete, the team must verify what Thunderbird deletes locally or remotely, how synchronised folders behave and whether a false positive can be reviewed. These are implementation and evaluation questions, not assumptions established by the policy table.

Table 4.2 Specified combined severity matrix

C1 / C3	Low	Medium	High	Critical
Low	Low	Medium	High	Critical
Medium	Medium	Medium	High	Critical
High	High	High	Critical	Critical
Critical	Critical	Critical	Critical	Critical

4.5 Team contribution and dependency matrix

The final contribution matrix must identify actual authors and distinguish component research, implementation, data preparation, integration, testing and writing. The supplied documents do not name the owners. Assigning an author to a component without their confirmation would misrepresent the project.

The dependency structure is clear at the design level: C2 requires bounded and interpretable C1/C3 results; the dashboard requires persisted events; the profiler requires recorded interactions; and enforcement requires extension permissions and a supported Thunderbird operation. Individual contribution declarations should refer to artefacts that verify this work.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The supplied material identifies Thunderbird as the client platform, DistilBERT for scam classification, distilgpt2-derived features for AI-writing assessment, a trained logistic combiner for scam scoring, a lightweight linear classifier for awareness selection and Flask for C3 image processing. These are design-level technology choices [1]–[4].

The final implementation description needs the exact package versions, model artefact identifiers, repository revision, operating system, hardware, runtime environments and storage mechanism. No source repository or deployment manifest was supplied in this task. This report therefore does not assert specific ports, installed versions, database tables or completed security controls from earlier project discussions.

5.2 Development and integration process

The report should describe actual development practice using dated evidence. Appropriate records include interface agreements, version-control changes, issue resolution, peer review and integration milestones. A narrative of sprints or continuous integration should be included only if the corresponding records exist.

A reproducible implementation record should connect each research objective to a design artefact, code change and validation result. Changes to the severity matrix, feature definitions or model calibration should be versioned because they alter observable behaviour. The training and evaluation pipelines must preserve dataset separation to prevent accidental leakage.

5.3 Deployment and operational configuration

The intended installation flow distributes the extension through an external website and requires the user to install and configure it in Thunderbird. The overview distinguishes selected-content analysis and optional protective-action permissions

and proposes a “Learn more” explanation for these requests [1]. The thesis should document the exact permission manifest and setup behaviour from the final build.

Deployment verification should demonstrate a fresh installation, a successful selected-message scan, recovery after backend failure and clear reporting when a model is unavailable. Backup and restoration procedures must account for configuration, retained event data and model artefacts. These procedures are proposed verification requirements because installation scripts and operational records were not supplied.

5.4 Evidence of team execution

The common report should synthesise team execution rather than paste raw logs. A concise evidence register can link planning decisions, meeting records, repository revisions and integration tests to the outcomes they support. Records from Planner, Teams or research diaries should be included only where they were actually used.

At present the evidence consists of four system-design sources and two academic structure or formatting sources [1]–[6]. They establish intended behaviour and reporting requirements. They do not establish who implemented each feature, when it was completed or whether the final extension enforced the required actions.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

Evaluation must address the complete workflow as well as individual classifiers. The main questions are whether the system identifies relevant threats, preserves separate scam and AI-writing interpretations, combines independent evidence correctly, executes the requested response and communicates uncertainty understandably. Resource cost and operational reliability must also be measured.

Text evaluation requires labelled legitimate and scam messages with documented provenance and appropriate separation by source or time. Image evaluation requires clean/stego pairs and separation by source image so derivatives do not leak across splits. AI-writing evaluation requires labels whose collection method is disclosed. Uncertain labels should not be converted into unquestioned ground truth.

The scam detector should report precision, recall, F1, false-positive rate and a confusion matrix at the operational threshold. Calibration should be evaluated separately using reliability plots and an appropriate probability-error metric. C3 should report performance by embedding method and payload setting. Overall enforcement tests must compare predicted policy, requested action and actual mailbox state.

6.2 Integration and end to end testing

The minimum integration suite should include a legitimate message without images, a text-only scam, clean text with a stego image, both analysis paths flagging, and multiple image attachments with mixed outcomes. Boundary tests must cover every severity transition, especially the High-plus-High escalation. The integer-band notation also requires a declared policy for fractional scores.

Negative cases should include a short message containing a malicious link, an unsupported image, a corrupt image, missing authentication context, a classifier timeout and an unavailable service. Permission-denied tests must show whether the

message remains accessible. Retry tests should ensure that repeating a scan does not duplicate an enforcement action or inflate the profile history.

Security testing should treat email bodies, headers, filenames, URLs and images as untrusted input. Dashboard rendering must be tested for unsafe HTML interpretation. Image-size limits and execution budgets should be exercised. No security-test pass rate is reported here because no execution evidence was supplied.

6.3 Overall results

The defensible result of the current document review is a synthesis of the intended architecture: three complementary components, separate scam and AI-writing outputs, independent image analysis and a fixed response-arbitration matrix. This is an architectural finding, not experimental validation.

No measured integrated results were supplied. Detection accuracy, false-positive reduction, calibration quality, latency, memory use, enforcement success and awareness improvement therefore remain unreported. Table 6.1 distinguishes specified behaviour from evidence required. The illustrative score of 55 and stego score of 80 in the source documents are examples of policy operation, not observations from a test dataset.

The final results section must add dataset sizes, hardware, model versions, experimental protocol, numerical outcomes and uncertainty. It should report unsuccessful cases and partial results with the same care as successes. Component-level figures must not be presented as end-to-end performance unless the same experiment actually measured the integrated system.

Table 6.1 Evaluation evidence available for this draft

Outcome	Current evidence	Required completion
Architecture	Design documents	Implementation traceability
Classification quality	No measured results supplied	Labelled test sets and confusion matrices
Calibration	Formula specified	Held-out reliability analysis
Enforcement	Actions specified	Actual Thunderbird outcome tests

Outcome	Current evidence	Required completion
Lightweight operation	Design intention	Hardware and resource benchmarks
Awareness benefit	Templates and profiler described	Participant study and limitations
Integration reliability	Workflow specified	Timeout, failure and retry results

6.4 Stakeholder and user validation

No participant study, interview transcript, ethics approval or user-acceptance record was supplied. Consequently, this draft does not claim that users found the interface intuitive or became safer. Such claims require a suitable study, informed participation and a documented collection process.

A useful validation study would ask participants to identify the reason for a warning, interpret uncertainty, recognise which action occurred and choose a safe next step. It should distinguish immediate comprehension from retained learning. Behavioural interaction logs alone cannot establish that users have become less vulnerable outside the tested environment.

6.5 Achievement of common objectives

The supplied architecture documents specify separate scam and AI-writing scores and a three-component workflow. They also specify severity fusion, template-based awareness and transparent profiling. These objectives are documented at design level. Their implementation and empirical success remain to be verified.

The evaluation should classify each objective as achieved, partly achieved or not achieved only after mapping it to evidence. A diagram can support architectural completeness; it cannot establish measured accuracy or enforcement. The final objectives matrix must retain links to test identifiers, result artefacts and code or model versions.

6.6 Integrated limitations and threats

The principal source limitation is inconsistency between documents. C1 depicts a whole-engine skip for short or previously legitimate messages, while C2 expects

independent text analysis. The C1 bundle includes external C3 evidence, whereas the overall design describes sibling results. C3 lists PNG and JPEG, but the system scope says PNG-only LSB detection. C2 alternates between evidence-driven reasoning and score-only action selection.

Validity threats include leakage across related messages or image derivatives, class imbalance, narrow embedding-method coverage, domain shift, uncertain AI-authorship labels and selection bias in user studies. Policy escalation is a design choice rather than a statistically validated consequence of independent findings. Operational independence also does not establish statistical independence of errors.

The profiler’s seven states depend on recorded interactions and sample size. The C2 redesign proposes at least three interactions per half, per-category safe-action rates, severity weighting and a broader Needs Support condition. Some thresholds are illustrative, and the complete final formula is absent. Those proposals cannot be described as a validated behavioural instrument.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

MailSentra's proposed value is a single understandable workflow for selected-message analysis, protective action and learning. Individuals may benefit from explanations connected to the email they are reviewing. Small organisations may value a clear installation and support model. These benefits remain hypotheses until stakeholder and deployment evidence is collected.

The affordability claim requires total-cost analysis covering hardware, configuration, updates, model maintenance, support and time lost to false positives. A freely available extension can still impose substantial operating costs. The report should therefore distinguish purchase price from sustained use.

7.2 Market alternatives and differentiation

A formal comparison should cover existing Thunderbird controls, email-security products, phishing-awareness tools and standalone steganalysis systems. Relevant comparison dimensions include deployment, supported content, explanation quality, enforcement, privacy, operating cost and reproducible evaluation. No current market study was provided.

The proposed differentiator is the connection between independent text and image assessments and evidence-linked user responses. This should be tested against actual alternatives using equivalent tasks. The thesis should not claim market uniqueness or superior accuracy from a feature list alone.

7.3 Business model and feasibility

Possible sustainability models include an openly maintained research release, a supported organisational deployment or a service-and-maintenance model. The project has not supplied a selected revenue model, pricing, partner agreements or operating budget. These possibilities should therefore remain options rather than asserted commercial commitments.

Technical feasibility depends on installation reliability, inference cost, classifier maintenance and compatibility with supported Thunderbird versions. Operational feasibility depends on who handles updates, incident review and false positives. A small pilot with documented support effort would provide stronger evidence than an assumed cost advantage.

7.4 Adoption intellectual property and compliance

Adoption barriers include manual setup, user trust, permission requests, model downloads and the consequences of blocking legitimate messages. The interface should explain selected-content access and optional protective actions in terms users can understand. The record should disclose what content is processed and retained.

Ownership and licensing must be checked separately for source code, pretrained weights, training datasets, images and third-party libraries. The team must also establish appropriate handling of personal email and behavioural interaction data. This draft does not provide a legal compliance determination; no completed licensing register, consent documentation or jurisdiction-specific review was supplied.

Permanent deletion deserves explicit evaluation because classification errors may have irreversible consequences. Restricted mode must also be evaluated against Thunderbird’s actual capabilities: rendering inert text inside the add-on does not by itself prove that every alternative client action is prevented. The thesis must describe the implemented enforcement boundary precisely.

7.5 Sustainable Development Goal contribution

The intended educational contribution should be evaluated through comprehension and retention, while infrastructure accessibility should be evaluated through deployment effort and resource measurements. Neither contribution follows automatically from publishing software. The final report should cite the selected SDG targets and explain the limited relationship supported by evidence.

Sustainability also involves maintaining threat feeds, retraining or recalibrating models when needed and keeping awareness explanations accurate. The team should consider energy and hardware costs, update availability and the burden on operators.

A maintenance plan is necessary to sustain any benefit beyond the research demonstration.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

MailSentra’s documented design connects selected-email text analysis, independent image steganalysis and a defense-and-awareness layer within a Thunderbird workflow. It preserves the distinction between scam risk and AI-writing assessment and proposes one response and explanation for combined findings. This provides a coherent basis for an integrated research prototype.

The current evidence establishes the design, not the operational effectiveness of the complete system. The final thesis must substantiate detection quality, calibration, resource use, enforcement and user outcomes through reproducible evaluation. Until that evidence is added, claims of reduced false positives, improved safety or affordability remain research objectives.

8.2 Principal integrated contribution

The principal contribution described by the sources is the relationship between separate evidence paths and an understandable response. C1 contributes scam indicators and an independently reported AI-writing assessment; C3 contributes image predictions and model visualisations; C2 connects findings with named actions, constrained awareness templates and an explicit interaction profile.

The integrated contribution is stronger when every result is traceable to its input, model and policy version and every action is traceable to actual enforcement. Establishing this traceability is essential for explaining failure, reviewing false positives and reproducing the research. It should be treated as a measurable integration outcome.

8.3 Future directions

The first priority is to resolve the documented inconsistencies and freeze an authoritative integration contract. The second is to provide implementation evidence and run the complete evaluation framework. Training provenance, leakage

prevention, calibration and realistic operational testing are necessary before extending the supported threat space.

Subsequent work can investigate broader image formats and embedding methods, robustness to new text-generation systems, accessible explanation design and longitudinal awareness evaluation. Audio, video and other email clients are outside the current scope. Any expansion should retain separate uncertainty reporting and be justified by measured benefit rather than the number of added features.

REFERENCES

[1] Project team, “Full system overview document(5),” unpublished project design document, supplied for this report, n.d.

[2] Project team, “Component 01 Architecture(3),” unpublished HTML architecture reference, supplied for this report, n.d.

[3] Project team, “Component 2 — Defense and Awareness: Complete Design Document,” Oct. 9, 2026. Unpublished project design document.

[4] Project team, “component 03 - system overview document(1),” unpublished project overview, supplied for this report, n.d.

[5] “Common Report Content,” supplied integrated-report template, n.d.

[6] Sri Lanka Institute of Information Technology, “Guidelines on Documentation and Submission of Dissertations,” supplied PDF, n.d.

[7] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” arXiv:1910.01108, 2019. https://arxiv.org/abs/1910.01108

[8] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-CAM: Visual Explanations From Deep Networks via Gradient-Based Localization,” Proceedings of ICCV, 2017. https://openaccess.thecvf.com/content_iccv_2017/html/Selvaraju_Grad-CAM_Visual_Explanations_ICCV_2017_paper.html

APPENDIX A SOURCE RECONCILIATION REGISTER

This register records questions that materially affect the final system description. A decision should be linked to the authoritative specification and verified implementation. It does not imply that a proposed correction has already been implemented.

Issue	Source difference	Required resolution
Trigger	Manual analysis versus arrival/opening narratives	Confirm selected-message invocation
Research components	Template has four; overview has three	Confirm team size and adapt headings
C3 routing	C1 bundle versus sibling results	Confirm integration ownership
Short text	Whole-engine bypass versus AI-writing gate	Define safe short-message treatment
Image scope	PNG-only versus PNG/JPEG	Confirm formats, sizes and embedding methods
C2 decision	Evidence-based wording versus score-only policy	Freeze action policy
Scores	Integer bands; C3 scale unspecified	Define fractional boundaries and normalisation
Heatmap	Visual proof wording versus model explanation	Use appropriate uncertainty language
Profiler	Existing formula versus proposed redesign	Provide final thresholds and code
Storage	Encrypted shared log asserted in design	Provide actual storage and security evidence

APPENDIX B INTEGRATED TEST MATRIX

The cases below are proposed tests. Status is not executed because no run evidence was supplied. Expected outcomes must be finalised against the approved policy and deployment configuration.

Test	Scenario	Expected verification
T01	Legitimate message, no image	Completed scan; quiet accounting
T02	Text scam only	C1 output and policy mapping
T03	Clean text, stego image	C3 runs independently of C1
T04	Both findings High	Combined Critical per matrix
T05	Multiple mixed images	Separate results; aggregation policy tested
T06	Short text, malicious URL	Skip behaviour does not hide analysis status
T07	Unsupported or corrupt image	Unsupported/failed status, not Clean
T08	One analysis path times out	Incomplete result and visible limitations
T09	Protective permission denied	Action failure accurately reported
T10	Quarantine and release	Verified folder state and review path
T11	Critical block on test mailbox	Exact deletion semantics documented
T12	Repeated request	No accidental duplicate action/profile entry
T13	Untrusted HTML or filename	Safe dashboard presentation
T14	Boundary and fractional scores	No unmapped values
T15	Inconsistent ML template choice	Deterministic grounded fallback
T16	Small interaction history	Honest insufficient-data state

APPENDIX C INFORMATION NEEDED FOR FINAL SUBMISSION

The report is a substantive research draft. Completing it requires author details, official degree information, confirmed component ownership, supervisor details, final policy decisions, actual deployment architecture and measured evaluation evidence. A full related-work review is also needed to support the research gap and comparative claims. These items cannot be replaced with illustrative examples.

The supplied formatting guideline specifies a minimum of 50 A4 pages [6]. This draft does not claim final compliance with that length requirement. The final report should reach the required extent through verified literature synthesis, implementation evidence, readable diagrams and actual experimental analysis rather than padding or fabricated results.

Before submission, the authors should verify every statement, complete the declaration, confirm the AI-assistance disclosure, add actual contributions, validate citations and update all Word fields. Any final classification of objectives as achieved must refer to evidence in Chapter 6.